Multi-Satellite Beam Hopping Based on Load Balancing and Interference Avoidance for NGSO Satellite Communication Systems

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***Abstract*— Due to the non-uniform distribution of the ground traffic demand and the high mobility of non-geostationary orbit (NGSO) satellites, how to make full use of the limited beam resources to serve users flexibly and efficiently is a brand-new challenge for NGSO communication systems. In order to achieve efficient spectrum utilization, the combination of full frequency multiplexing and beam hopping is a major trend in future satellite communication systems. However, conventional beam hopping methods are mainly based on geostationary satellites, which do not take into account the interference between satellites. This paper proposes a multi-satellite beam hopping algorithm based on load balancing and interference avoidance, which takes advantage of the multiple coverage features in the NGSO constellation and avoids intra-satellite interference and inter-satellite interference by designing beam-hopping patterns with spatial isolation characteristics. In particular, we decompose the multi-satellite beam hopping problem into three sub-problems, which are the multi-satellite load balancing problem, the single-satellite beam hopping pattern design problem, and the multi-satellite interference avoidance problem. Simulation results demonstrate that the proposed method reduces the load gap among satellites by about 72.5% and the average traffic satisfaction rate can reach 81.4%. Besides, our method has the lowest unmet capacity compared with other benchmarks, achieving better offered-requested data match.**

***Index Terms*— Multi-satellite beam hopping, resource allocation, load balancing, interference avoidance, NGSO satellite communications.**

I. Introduction

SATELLITE communication systems have the advantage of wide coverage and low construction cost, which is the trend of next-generation communication systems [1], [2], [3], [4]. Recently, non-geostationary orbit (NGSO) constellations have developed rapidly, such as Oneweb, Starlink, Telesat, etc [5]. In order to provide broadband transmission and seamless coverage, the NGSO constellation needs to achieve multiple coverage, that is, some areas are covered by multiple satellites. In addition, due to the non-uniform geographic distribution of the ground users and high mobility of NGSO satellites, the traffic demands in the area seen by the satellite show temporal and spatial non-uniformity. Therefore, how to make full use of the feature of multiple coverage to match limited satellite resources to non-uniform services is a major challenge for NGSO satellite systems [6], [7].

The multi-beam satellites (MBS) play an important role to solve above problem, which can generate a large number of beams simultaneously and flexibly allocate these beam resources to improve system efficiency. Some MBS-based resource allocation algorithms have been proposed from multiple dimensions such as bandwidth and power [7], [8], [9], [10], [11]. Aravanis et al. [10] proposed a two stage genetic algorithm and simulated annealing (GA-SA) to allocate beam power, while [12] presented a modified SA method to achieve a joint bandwidth and power allocation in MBS. In order to reduce the computationally complex of iterative meta-heuristics, [13] proposed a dynamic channel allocation with a deep reinforcement learning-based framework. However, above works are mainly based on fixed MBS, where the illumination direction of beams can not be changed. Thus, the space freedom of the beam is not fully utilized.

In recent years, beam hopping (BH) technology has been widely applied in MBS to fully explore the freedom of time and space freedom of beams [14], [15], [16]. Furthermore, in order to support BH in MBS, several super-framing specifications have been proposed in DVB-S2X standard [17]. In BH systems, there are K beams serving N ($N \gg K$) cells in a time division manner. Besides, the illumination direction of activated beams, which is called the BH pattern, varies in each time slot. Therefore, the big challenge in BH systems is that how to design the BH pattern to match the non-uniform traffic demand. References [18], [19] conducted a series of research work to show the advantages of the highly flexible dynamic beam hopping (DBH) method in MBS. A heuristic algorithm based on greedy method was proposed in [20] to improve the capacity of the BH system while [21]

Manuscript received 12 April 2022; revised 23 August 2022 and 23 October 2022; accepted 28 November 2022. Date of publication 1 December 2022; date of current version 16 January 2023. This work was supported by the National Key Research and Development Program under Grant 2019YFB1803000. The associate editor coordinating the review of this article and approving it for publication was S. Coleri. *(Corresponding author: Zuyao Ni.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TCOMM.2022.3226190.

Digital Object Identifier 10.1109/TCOMM.2022.3226190

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proposed a beam hopping method based on Max-Min rate control. In order to take the system throughput into account while guarantying the delay requirements of terminals, [22] presented a multi-objective optimization problem and adopted heuristic method to achieve a joint power and bandwidth allocation for BH. To further improve spectrum utilization, full frequency multiplexing combined with precoding was proposed in [23], [24], and [25]. However, accurate acquisition of channel information is required when using precoding. With the development of deep learning (DL), DL has been viewed as a promising method to deal with complex resource allocation in BH systems [26], [27], [28]. A combined DL-and-optimization approach was proposed in [26] to achieve a rapid and feasible BH scheduling, but this method employs supervised learning and needs a large number of labels generated by SA or GA. References [27], [28] viewed DBH as a sequential decision-making problem and proposed a deep reinforcement learning (DRL) method to deal with a multi-objective Markov Decision Process (MDP) for BH.

However, the above-mentioned researches mainly focus on the single satellite, especially on GEO. In the NGSO satellite communication scenario, due to the small size of the terminal and the characteristics of multiple coverage, terminals will be more susceptible to interference from neighboring NGSO satellites. Therefore, in the NGSO multi-satellite beam-hopping scenario, not only the on-demand allocation of beam resources for each satellite must be considered, but also the interference avoidance between satellites should be reasonably designed to improve the communication efficiency of the entire system. In other words, the above works cannot be directly transplanted to NGSO multi-satellite beam hopping. Within the limited scope of the authors' knowledge, researches on multi-satellite beam hopping have not been conducted.

To deal with the above problems, this paper proposes a multi-satellite beam hopping method based on load balancing and interference avoidance. On the one hand, we balance traffic load of satellites by assigning cells to different satellites, thereby reducing the load burden on each satellite; on the other hand, to avoid both intra-satellite co-frequency interference and inter-satellite co-frequency interference, this paper selects BH patterns which meet the spatial isolation condition and performs time slot scheduling on them, satisfying the traffic demand of each cell as much as possible. Simulation results show that the proposed method can better match non-uniform traffic demand with limited resources. The main contributions and innovations of this work are summarized as follows.

- This paper first constructs a multi-satellite beam hopping optimization problem for efficient resource utilization. The original NP-hard and non-convex optimization problem is decomposed into three simple sub-problems, namely the multi-satellite load balancing problem, the single-satellite beam hopping pattern design problem, and the multi-satellite interference avoidance problem.
- Exploiting the multiple coverage characteristics of NGSO satellite system, this paper balances the satellite load, so that the traffic demand of each satellite is as equal as possible, improving the efficiency of the entire satellite communication system. Furthermore, a greedy-based load balancing scheme is proposed to reduce the computational complexity.
- In order to reduce the intra-satellite interference and realize on-demand transmission, BH patterns with spatial isolation characteristics are selected by each satellite independently, which is formulated as a linear integer optimization problem.
- For the purpose of further reducing the co-frequency interference between satellites, this paper establishes multi-satellite interference avoidance problem by arranging the order of satellites' BH patterns. Besides, a low-complexity heuristic algorithm is devised to solve this problem.

TABLE I
NOTATIONS AND DEFINITIONS

Notations	Definitions
N_s	Number of satellites
N_c	Number of cells covered by N_s satellites
N	Number of cells covered by each satellite
K	Number of beams simultaneously generated by each satellite
V_i	The set of cells covered by the i-th satellite
T_H	The period of beam hopping time plan
T_{slot}	The duration of each time slot
N_{slot}	Number of slots that one BH period contains
$x_{i,n}^t$	Indication whether the cell n of satellite i is illuminated at slot t
P_k	The power of beam k
P_{sat}	The total satellite power
$h_{k,l}$	The channel coefficient between the k-th user and the l-th beam
B	Total satellite bandwidth
T_{rx}	The noise temperature of the receiver
G_r^k	The k-th users receive antenna gain
$G_t(\theta_{k,l})$	The transmit antenna gain from the l-th beam to the k-th user
$\theta_{k,l}$	The off-axis angle between the user k and the main lobe of beam l
d_n	The traffic demand rate of cell n
D_n	The total traffic demand of cell n during one BH period
r_n^t	The channel capacity of the cell n at time slot t
C_n^t	The offered traffic of the cell n at time slot t
R_n	The total offered traffic of the cell n during one BH period
ψ_i	The set of beams belonging to satellite i
S_i	The traffic load of satellite i
$u_{i,n}$	Indication whether the cell n is assigned to satellite i
N_i	Number of cells allocated to satellite i
M_i	Number of activated beams for satellite i
N_v^i	Number of valid BH patterns for satellite i
$\boldsymbol{v}_m^i$	m-th valid BH pattern of satellite i
$\boldsymbol{c}_m^i$	Service vector of m-th valid BH pattern for satellite i
ϕ_m^i	Number of times that the i-th satellite selects m-th valid BH pattern
t_{mn}^i	Indication whether m-th valid BH pattern of satellite i is used at slot n
β_n	Traffic satisfaction rate of cell n
$\hat{\beta}_i$	Average traffic satisfaction rate of satellite i
C_{unmet}^i	The unmet capacity of satellite i
C_{unused}^i	The unused capacity of satellite i

The rest of the paper is organized as follows. Section II describes the system model and formulates the multi-satellite beam hopping optimization problem. In section III, the optimization problem is decomposed into three sub-problems, and the solutions to these three problems are elaborated. Simulations prove that the proposed method can achieve superior performance in Section IV. Section V concludes this paper.

II. System Model and Problem Formulation

A. System Model

In this paper, we take the forward link of the multi-beam NGSO satellite communication system into account as shown

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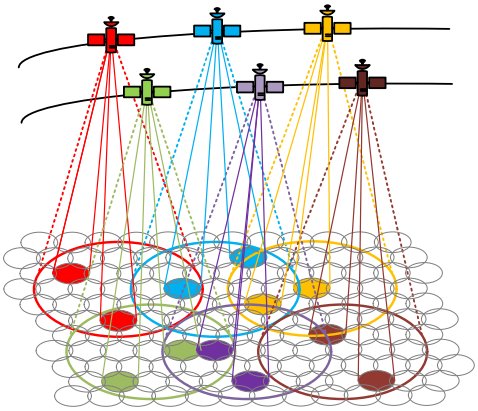


Fig. 1. Forward link of the multi-beam NGSO satellite communication system.

in Fig. 1. The detailed notations and definitions used in this paper are summarized in Table I. There are N_s satellites to serve a specific area and each satellite can cover N virtual cells. It should be noted that some virtual cells are covered by multiple satellites, which is a major feature for the NGSO communication system. Assuming that there are a total of N_c virtual cells covered by N_s satellites, the set of virtual cells covered by the i-th satellite is defined as V_i. Obviously, the size of each set $|V_i|$ is equal to N and the union size of all sets $\left|\bigcup_{i=1}^{N_s} V_i\right|$ is equal to N_c. Due to the non-uniform geographic distribution and time-varying characteristics of the users, the traffic demand of each virtual cell is quite different, which is denoted by d_n (measured by bps).

For beam hopping system, each satellite can simultaneously generate K ($K \ll N$) beams at most to serve these virtual cells in the time division multiplex (TDM) manner. The downlink beams are controlled by the satellite according to beam hopping time plan (BHTP), also called beam hopping pattern, as shown in Fig. 2. Generally, the period of BHTP is T_H, which contains N_{slot} time slots. The duration of each time slot is $T_{slot} = T_H/N_{slot}$, representing the shortest dwell time of each beam.

In order to meet the traffic demand of virtual cells, satellites should reasonably determine the irradiation position of the beams (i.e., which K virtual cells should be served at each time slot). Therefore, the BHTP of the i-th satellite can be represented as follows,

$$\boldsymbol{X}_i = \left[\boldsymbol{x}_i^1, \boldsymbol{x}_i^2, \ldots, \boldsymbol{x}_i^t, \ldots, \boldsymbol{x}_i^{N_{slot}}\right], \tag{1}$$

where $\boldsymbol{x}_i^t$ denotes the pattern of the i-th satellite at time slot t and it can be expressed as $\boldsymbol{x}_i^t = [x_{i,1}^t, x_{i,2}^t, \ldots x_{i,n}^t, \ldots, x_{i,N}^t]^T$. $x_{i,n}^t \in \{0,1\}$ indicates that whether the virtual cell n of the satellite i is illuminated by the beam at time slot t.

In order to make full use of spectrum resources and improve spectrum utilization, we assume that all beams occupy the entire satellite bandwidth B. In other words, full frequency reuse is applied to the multi-satellite beam hopping system and the co-frequency interference between beams cannot be ignored.

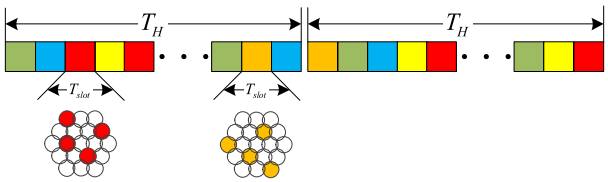


Fig. 2. Illustration of beam hopping time plan.

In the forward link of multi-satellite beam hopping system, each satellite generate K beams to serve ground users from different cells, and the ground user has one antenna. Taking the user served by the k-th beam (also called the k-th user) as an example, the one will receive N_sK signals from N_s satellites, and the channel vector between this user and N_sK beams can be denoted by $\boldsymbol{h}_k = [h_{k,1}, h_{k,2}, \ldots, h_{k,N_sK}] \in \mathbb{C}^{N_sK\times 1}$. Assuming that the Doppler frequency offset caused by satellite motion is compensated by the user and the sky is clear so that rain attenuation is not considered, the channel coefficient between the k-th user and the l-th beam, similar to the expression in [25], can be further denoted by,

$$h_{k,l} = \frac{\sqrt{G_t(\theta_{k,l})G_r^k}}{4\pi\frac{d_{k,l}}{\lambda}}, \tag{2}$$

where $G_t(\theta_{k,l})$ represents the transmit antenna gain from the l-th beam to the k-th user and it is determined by the off-axis angle $\theta_{k,l}$ between the user k and the main lobe direction of beam l. G_r^k denotes the receive antenna gain of the user k. $d_{k,l}$ is the distance between the user k and the beam l, and λ refers to the wavelength.

As a result, the signal to interference plus noise (SINR) of the user k can be expressed as

$$SINR_t^k = \frac{P_k\left|h_{k,k}\right|^2}{k_BT_{rx}B + \underbrace{\sum_{l\neq k, S(l)=S(k)} P_l\left|h_{k,l}\right|^2}_{\text{Intra-satellite interference}} + \underbrace{\sum_{l\neq k, S(l)\neq S(k)} P_l\left|h_{k,l}\right|^2}_{\text{Inter-satellite interference}}}, \tag{3}$$

where k_B denotes the Boltzmann constant and T_{rx} refers to the receiver noise temperature. $S(k)$ denotes the satellite where beam k belongs and P_k represents the transmit power of beam k. It is noted that the co-frequency interference consists of two parts. One is the intra-satellite interference caused by beams which are in the same satellite and the other is the inter-satellite interference from other satellites' beams.

At time slot t, if the virtual cell n is illuminated by the beam k, the offered traffic (available throughput) of cell n can be calculated as

$$r_n^t = B\log_2(1 + SINR_t^k). \tag{4}$$

Conversely, if cell n is not served in the time slot t, the offered traffic $r_n^t = 0$.

B. Optimization Problem Formulation

In the beam hopping system, beams serve different cells at different time slot in order to meet the non-uniform traffic

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demand as much as possible with limited beam resources. The traffic demand of cell n in one BH period T_H can be defined as $D_n = d_n \times T_H$. In addition, the offered traffic amount of cell n at time slot t can be expressed as $C_n^t = r_n^t \times T_{slot}$.

In order to measure the quality of service, we define the traffic satisfaction rate for each cell, which is denoted by the ratio of offered traffic against the demand traffic (i.e. $\frac{R_n}{D_n}$) during the whole BH period. The optimization goal of multi-satellite beam hopping is to maximize the minimum traffic satisfaction rate among all cells, which means that the traffic demands of each cell need to be satisfied as much as possible. In other words, the cells with low demands also need to be served, otherwise the objective function will be reduced. Therefore, the fairness among cells is guaranteed. The considered optimization problem is detailed below as,

$$
\begin{aligned}
P_0: \underset{\boldsymbol{X}_1,\boldsymbol{X}_2,\ldots,\boldsymbol{X}_{N_s}}{\text{maxmize}} \ & \min_n \frac{R_n}{D_n} \\
\text{s.t. } C_1: & \sum_{n=1}^{N} x_{i,n}^t \leqslant K, \quad t=1,\ldots,N_{slot}; i=1\ldots,N_s, \\
C_2: & R_n = \sum_{t=1}^{N_{slot}} C_n^t, \quad n=1,\ldots,N_c, \\
C_3: & x_{i,n}^t \in \{0,1\}, \quad i=1\ldots,N_s; n=1,\ldots,N, \\
C_4: & \sum_{k\in\psi(i)} P_k \leqslant P_{sat}, \quad i=1\ldots,N_s.
\end{aligned} \tag{5}
$$

Matrices $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_{N_s}$ are binary variables to be optimized, which are defined as the BHTP of the satellites. C_1 indicates that each satellite can generate up to K beams and C_2 defines the offered traffic in one BH period. C_3 constrain optimization variables to binary variables and C_4 requires the sum of beam power is less than the total satellite power P_{sat}, and ψ_i denotes the set of beams belonging to satellite i.

Lemma 1: P_0 is NP-hard.

Proof: Please refer to Appendix A. □

Being aware of the non-convexity and NP-hardness of P_0, it is not feasible to solve P_0 directly. Thus, we decompose P_0 into three sub-problems which are easy to solve, and get a sub-optimal solution, which will be further described in next section.

III. Load Balancing and Interference Avoidance for Multi-Satellite Beam Hopping

This section divides P_0 into three sub-problems, which are the multi-satellite load balancing problem, the single-satellite beam hopping pattern design problem, and the multi-satellite interference avoidance problem. The whole structure is shown in Fig. 3.

A. Multi-Satellite Load Balancing Problem

In the NGSO beam hopping system, the visual traffic demand of each satellite is quite different due to the uneven distribution of terrestrial traffic. As shown in Fig. 4, some satellites cover densely populated areas while others cover less populated area. If these adjacent hotspot cells are all assigned to the nearest satellite, this satellite will be overburdened while

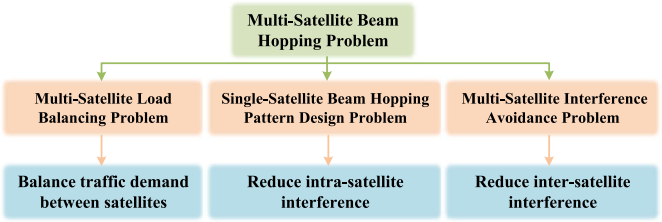


Fig. 3. Multi-satellite beam hopping problem decomposition architecture.

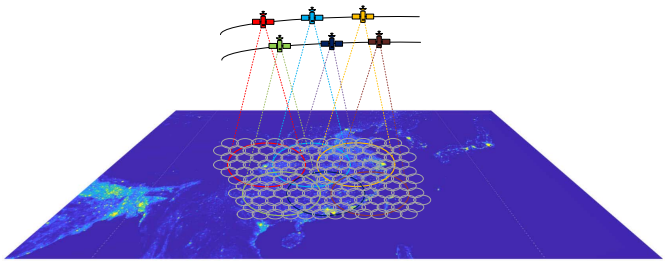


Fig. 4. Illustration of uneven traffic demand between NGSO satellites.

the resources of other satellites will be wasted. Therefore, achieving load balancing between satellites is of great importance to improve the resource utilization of the entire system.

Considering that some hotspot cells can be covered by multiple satellites, we can assign them to different satellites, thereby reducing the traffic burden of a single satellite. Denote S_i as the traffic load of i-th satellite. The goal of multi-satellite load balancing problem is to minimize the traffic load gap between satellites by properly allocating cells to different satellites. The optimization problem can be further described as follow,

$$
\begin{aligned}
P_1: \underset{\boldsymbol{u}_1,\boldsymbol{u}_2,\ldots,\boldsymbol{u}_{N_s}}{\text{minimize}} \ & \{\max_i\{S_i\} - \min_i\{S_i\}\} \\
\text{s.t. } C_1: & S_i = \boldsymbol{d}^T\boldsymbol{u}_i = \sum_n d_n u_{i,n}, \quad i=1\ldots,N_s \\
C_2: & u_{i,n} = 0, \quad \forall n \notin V_i; i=1,\ldots,N_s, \\
C_3: & \sum_i u_{i,n} = 1, \quad n=1,\ldots,N_c, \\
C_4: & u_{i,n} \in \{0,1\}, \quad i=1\ldots,N_s; n=1,\ldots,N_c.
\end{aligned} \tag{6}
$$

Vectors $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{N_s}$ with the size of $N_c \times 1$ are the binary variables to be optimized, which are defined as cell-satellite allocation vector. $u_{i,n}$ is the n-th element in $\boldsymbol{u}_i$ and $u_{i,n} = 1$ means the cell n is served by satellite i in the whole BH period. C_1 defines the total traffic load that each satellite needs to serve and C_2 indicates that cells can only be allocated to satellites that they can see and V_i represents the set of cells covered by the i-th satellite. C_3 requires that each cell can only be served by one satellite during the BH period. For convenience, we introduce auxiliary variables $\vartheta \triangleq \max\{S_i\}$ and $\kappa \triangleq \min\{S_i\}$. Rewrite P_1 as follow,

$$
\begin{aligned}
P_2: \underset{\boldsymbol{u}_1,\boldsymbol{u}_2,\ldots,\boldsymbol{u}_{N_s}}{\text{minimize}} \ & \vartheta - \kappa \\
\text{s.t. } C_1: & S_i = \boldsymbol{d}^T\boldsymbol{u}_i = \sum_n d_n u_{i,n}, \quad i=1\ldots,N_s
\end{aligned}
$$

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$$
\begin{aligned}
&C_2: u_{i,n} = 0, \quad \forall n \notin V_i; i = 1, \ldots, N_s, \\
&C_3: \sum_i u_{i,n} = 1, \quad n = 1, \ldots, N_c, \\
&C_4: u_{i,n} \in \{0, 1\}, \quad i = 1 \ldots, N_s; n = 1, \ldots, N_c, \\
&C_5: S_i \leqslant \vartheta, \quad i = 1 \ldots, N_s, \\
&C_6: S_i \geqslant \kappa, \quad i = 1 \ldots, N_s.
\end{aligned} \tag{7}
$$

It can be noted that P_2 is a mixed integer linear programming (MILP) due to linear objectives and constraints, which is widely solved by branch and bound method [29]. This method can obtain globally optimum solution and can be effectively executed by MOSEK solver in convex optimization toolbox CVX [30].

In order to reduce the computational cost, this paper proposes a greedy-based load balancing scheme to obtain the a suboptimal solution to P_2, as shown in Algorithm 1. Firstly, we initialize all satellites' traffic load and sort all cells in descending order in terms of their demands. Then, traverse each cell according to this order, and assign the cell to the visible satellite with minimum traffic load.

Algorithm 1 A Greedy-Based Load Balancing Scheme

Input: The traffic demand of each cell d_n, the set of cells covered by the i-th satellite V_i, the number of satellites N_s, the number of cells N_c.

Initialize: $S_i = 0, i = 1, \ldots, N_s$.

1: Sort cells in descending order according to d_n: $(n_1, n_2, \ldots, n_{N_c})|_{d_{n_1} \geqslant d_{n_2} \geqslant \ldots \geqslant d_{n_{N_c}}}$.
2: **for** $n = n_1$ to n_{N_c} **do**
3: Obtain the set of visible satellites Θ_n for cell n: $\Theta_n = \{i | n \in V_i\}$.
4: Find the satellite i^* with minimum traffic load from Θ_n: $i^* = \arg\min_{i \in \Theta_n} S_i$.
5: Assign the cell n to the satellite i^*: $u_{i^*,n} = 1$.
6: Update the traffic load of satellite i^*: $S_{i^*} = S_{i^*} + d_n$.
7: **end for**

Output: The cell-satellite allocation vector $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{N_s}$.

B. Single-Satellite Beam Hopping Pattern Design Problem

After solving the problem P_2, the cells that each satellite needs to serve is determined, and each satellite should design beam hopping pattern to meet the traffic demand of cells. Assuming that there are N_i (obviously, $N_i \leqslant N$) cells allocated to satellite i for serving, there may be situations where N_i is close to K (i.e. the maximum number of beams for each satellite) or even less than K. Fig. 5 shows an example where $N = 19$, $N_i = 7$, and $K = 4$. It can be seen that the irradiation direction of beams are very close to each other, which causes serious co-channel interference. In addition, if $N_i < K$, the number of activated beams (defined as M_i) must be less than K to avoid wasting power. Thus, in order to reduce co-channel interference and avoid wasting resources, M_i should be adaptively changed according to N_i. Furthermore, M_i cannot be too small, otherwise all cells cannot be fully served. Based on the above considerations,

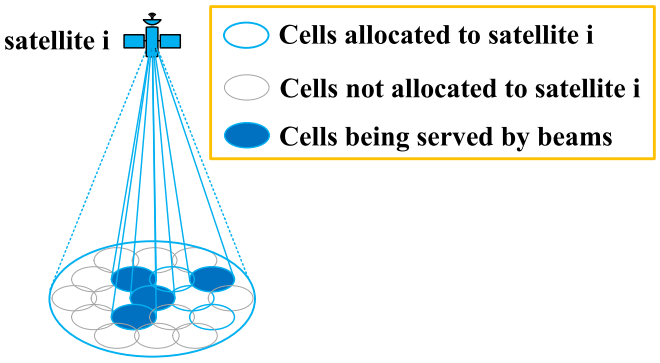


Fig. 5. Example of cells allocated to satellite i when $N = 19$, $N_i = 7$, and $K = 4$.

this paper sets M_i to be one quarter of N_i. As a result, we determine the value of M_i according to the principle as follow,

$$
M_i = \begin{cases} K, & N_i \geq 4\ K, \\ \lceil N_i/4 \rceil, & N_i < 4\ K, \end{cases} \tag{8}
$$

where $\lceil \cdot \rceil$ is the ceil function. It can be observed that the number of activated beams decreases as the number of cells decreases, so that the interference between beams can be reduced as much as possible through spatial isolation. Meanwhile, this principle applies to all types of NGSO satellites (MEO and LEO satellites) because this criterion is only related to the number of cells N_i, not the satellite altitude.

For the convenience of analysis, we take the beam hopping pattern design of the i-th satellite as an example. At each time slot, there are $C_{N_i}^{M_i}$ beam irradiation combinations, but not all illumination patterns can satisfy the spatial isolation between beams. Therefore, in order to reduce inter-beam interference, it is necessary to select beam patterns with spatial isolation conditions from all illumination patterns. Inspired by [23], we firstly define the adjacency matrix between cells $\boldsymbol{A} \in \mathbb{R}^{N_i \times N_i}$ and the element $a_{ij} \in \{0, 1\}$ represents whether the cell i is adjacent to cell j. In particular, a_{ij} can be further expressed as follow,

$$
a_{ij} = \begin{cases} 0, & i = j, \\ 1, & i \neq j, dist(i, j) \leq 2R, \\ 0, & i \neq j, dist(i, j) > 2R, \end{cases} \tag{9}
$$

where $dist(i, j)$ denotes the center distance between the cell i and the cell j; R denotes the cell radius.

Then, we can define a binary vector $\boldsymbol{z} \in \{0, 1\}^{N_i \times 1}$ to represent one BH pattern, where the locations of 1's in $\boldsymbol{z}$ indicate which cells are illuminated by beams, and the number of 1's in $\boldsymbol{z}$ must equal to M_i. According to the graph theory, the BH pattern with spatial isolation condition (also called the valid BH pattern) should meet the constraint that the irradiated cells are not adjacent to each other, which can be expressed as,

$$
\boldsymbol{z}^T \boldsymbol{A} \boldsymbol{z} = 0. \tag{10}
$$

Obviously, the number of valid BH patterns is much less than the total number of patterns $C_{N_i}^{M_i}$, which can be denoted by N_v^i. According to each valid BH pattern, we can calculate

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the service capacity of each cell from Eq. (4). In other words, each valid BH pattern corresponds to a service vector $\boldsymbol{c} \in \mathbb{R}^{N_i \times 1}$. Taking the m-th valid BH pattern of i-th satellite $\boldsymbol{v}_m^i$ (i.e., ${\boldsymbol{v}_m^i}^T \boldsymbol{A} \boldsymbol{v}_m^i = 0$) as an example, the corresponding service capacity vector $\boldsymbol{c}_m^i$ can be expressed as follow,

$$c_{mj}^i = \begin{cases} 0, & v_{mj}^i = 0, \\ r_j \times T_{slot}, & v_{mj}^i = 1, \end{cases} \tag{11}$$

where v_{mj}^i and c_{mj}^i are the j-th element of $\boldsymbol{v}_m^i$ and $\boldsymbol{c}_m^i$ respectively; r_j denotes the offered traffic for cell j as defined in Eq. (4). All service vectors can form a service matrix, which is denoted by $\boldsymbol{C}_i = [\boldsymbol{c}_1^i, \boldsymbol{c}_2^i, \ldots, \boldsymbol{c}_{N_v^i}^i] \in \mathbb{R}^{N_i \times N_v^i}$. A detailed description of selecting spatial isolation patterns is shown in Algorithm 2. Firstly, according to the coordinates of N_i cells, we calculate the distance between any two cells and obtain the adjacency matrix $\boldsymbol{A}$. Then, we find out valid BH patterns that satisfies Eq. (10) from a total of $C_{N_i}^{M_i}$ BH patterns. Finally, each valid BH pattern corresponds to a service vector, which can be obtained by Eq. (11). It should be noted that the complexity of Algorithm 2 depends on $C_{N_i}^{M_i}$.

Algorithm 2 Pseudo Code of Selecting Spatial Isolation Patterns for Satellite i

Input: The number of allocated cells for satellite i: N_i; the number of activated beams for satellite i: M_i; cell coordinates $Cell(lat, lon)$.

Initialize: $m = 0$.

1: Calculate the distance between any two cells and get the adjacency matrix $\boldsymbol{A}$ according to Eq. (9).
2: **for** $k = 1$ to $C_{N_i}^{M_i}$ **do**
3: Obtain the k-th beam hopping pattern $\boldsymbol{z}_k$.
4: **if** $\boldsymbol{z}_k^T A \boldsymbol{z}_k = 0$ **then**
5: $m = m + 1$.
6: Obtain the j-th valid beam hopping pattern $\boldsymbol{v}_m^i = \boldsymbol{z}_k$.
7: Calculate the corresponding service vector $\boldsymbol{c}_m^i$ according to Eq. (11).
8: **else**
9: $\boldsymbol{z}_k$ is not a valid beam hopping pattern.
10: **end if**
11: **end for**

Output: The number of valid patterns $N_v^i = m$ and the service matrix $\boldsymbol{C}_i = [\boldsymbol{c}_1^i, \boldsymbol{c}_2^i, \ldots, \boldsymbol{c}_{N_v}^i]$.

At each BH time slot, the satellite should choose one of the valid BH patterns to serve the allocated cells, and the selections during the whole BH period constitute the beam hopping time plan. The goal of single-satellite beam hopping pattern design problem is to choose reasonable valid BH patterns to satisfy the traffic demands of cells as much as possible. Assuming that ϕ_m^i denotes the number of times that the i-th satellite selects m-th valid BH pattern in the entire BH period, the optimization problem of the i-th satellite beam hopping pattern design can be expressed as

$$\begin{aligned} P_3: \underset{\phi_1, \phi_2, \ldots, \phi_{N_v^i}}{\text{maxmize}} \ & \min_n \frac{R_n}{D_n} \\ \text{s.t. } C_1: & \sum_{m=1}^{N_v^i} \phi_m^i = N_{slot}, \\ C_2: & R_n = \sum_{m=1}^{N_v^i} c_{mn}^i \phi_m^i, \quad n = 1, \ldots, N_i, \\ C_3: & \phi_m^i \in \mathbb{N}, \quad m = 1 \ldots, N_v^i, \end{aligned} \tag{12}$$

where c_{mn}^i denotes the offered traffic for the cell n of the m-th valid BH pattern (i.e. the element in the m-th row and n-th column of service matrix $\boldsymbol{C}_i$). $\phi_1^i, \phi_2^i, \ldots, \phi_{N_v^i}^i$ are the non-negative integer variables to be optimized and the optimization goal of P_2 is to maximize the minimum ratio of the offered traffic against the demand traffic among allocated cells. C_1 indicates that the sum of the occurrences of all valid BH patterns is equal to the number of time slots contained in one BH period. C_2 calculates the offered traffic of cell n in one BH period.

In order to better solve the problem P_3, we define an auxiliary variable $\xi \triangleq \min_n \frac{R_n}{D_n}$. Thus a new constraint should be introduced as $\frac{R_n}{D_n} \geqslant \xi, n = 1, \ldots, N_i$, and the problem P_3 can be renewed as follow,

$$\begin{aligned} P_4: \underset{\phi_1, \phi_2, \ldots, \phi_{N_v^i}, \xi}{\text{maxmize}} \ & \xi \\ \text{s.t. } C_1: & \sum_{m=1}^{N_v^i} \phi_m^i = N_{slot}, \\ C_2: & R_n = \sum_{m=1}^{N_v^i} c_{mn}^i \phi_m^i, \quad n = 1, \ldots, N_i, \\ C_3: & \frac{R_n}{D_n} \geqslant \xi, \quad n = 1, \ldots, N_i, \\ C_4: & \phi_m^i \in \mathbb{N}, \quad m = 1 \ldots, N_v^i. \end{aligned} \tag{13}$$

It can be noted that the problem P_4 is also a MILP, which can be solved by the classical branch and bound method [29] using MOSEK solver to provide globally accurate solution.

C. Multi-Satellite Interference Avoidance Problem

By solving problem P_4, each satellite can choose the valid BH pattern independently, which effectively reduces intra-satellite interference. However, the inter-satellite interference is ignored when solving problem P_4. Considering that P_4 only solve the number of occurrences of each valid pattern, and did not obtain the time sequence of these patterns, we need to arrange the pattern order of each satellite to achieve multi-satellite interference avoidance. In other words, at each time slot, the patterns of each satellite are as spatially isolated as possible. Therefore, we define a metric $\rho_{i,j}$ to measure the degree of isolation between patterns as follow,

$$\rho_{i,j} = \min_{p \in G_i, q \in G_j} \{dist(p, q)\}, \tag{14}$$

where $dist(p, q)$ denotes the distance between the cell p and the cell q; G_i and G_j represent the illuminated cells in the pattern i and the pattern j respectively. It can be seen that $\rho_{i,j}$ indicates the shortest distance between the served cells in the two patterns. As shown in Fig. 6, the larger the value, the better

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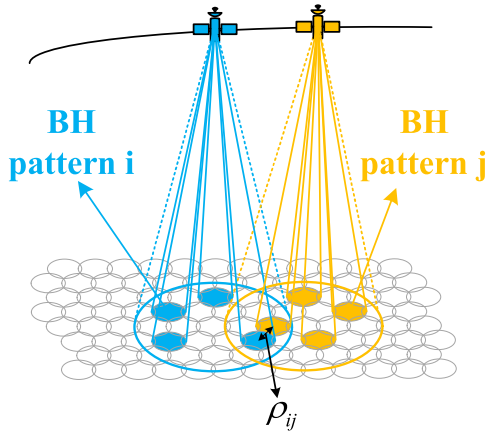


Fig. 6. Illustration of the spatial isolation between two beam hopping patterns.

the spatial isolation effect. On the contrary, enlightened by [25], we define a interference penalty $\omega_{i,j} = \frac{1}{\rho_{i,j}}$ to represent the mutual interference of two patterns.

Furthermore, we can define the interference penalty matrix $\Omega_{m,n} \in \mathbb{R}^{N_v^m \times N_v^n}$ for the valid BH patterns of satellite m and satellite n, which has the form as,

$$\Omega_{m,n} = \begin{bmatrix} \omega_{1,1} & \omega_{1,2} & \cdots & \omega_{1,N_v^n} \\ \omega_{2,1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \omega_{N_v^m-1,N_v^n} \\ \omega_{N_v^m,1} & \cdots & \omega_{N_v^m,N_v^n-1} & \omega_{N_v^m,N_v^n} \end{bmatrix}. \quad (15)$$

Meanwhile, we use $\boldsymbol{\Omega}$ to represent the interference penalty matrix for all satellites' valid BH patterns, which can be further expressed as

$$\boldsymbol{\Omega} = \begin{bmatrix} 0_{N_v^1 \times N_v^1} & \Omega_{1,2} & \cdots & \Omega_{1,N_s} \\ \Omega_{2,1} & 0_{N_v^2 \times N_v^2} & \ddots & \Omega_{2,N_s} \\ \vdots & \ddots & \ddots & \Omega_{N_s-1,N_s} \\ \Omega_{N_s,1} & \cdots & \Omega_{N_s,N_s-1} & 0_{N_v^{N_s} \times N_v^{N_s}} \end{bmatrix}. \quad (16)$$

After solving P_4, the number of occurrences of each valid pattern can be obtained, which is denoted by $[\phi_1^i, \phi_2^i, \ldots, \phi_{N_v^i}^i]$. In order to represent the chronological order of these valid patterns, we define a BH time slot schedule matrix $\boldsymbol{T}^i \in \{0,1\}^{N_v^i \times N_{slot}}$ for satellite i, where the element $t_{mn}^i = 1$ indicates that the m-th valid BH pattern is adopted at time slot n. $\boldsymbol{T} = [\boldsymbol{T}^1; \boldsymbol{T}^2; \ldots; \boldsymbol{T}^{N_s}]$ denotes the BH time slot schedule matrix of all satellites. The target of multi-satellite interference avoidance is to reduce inter-satellite interference by arranging the order of satellite patterns. Alternatively, we can use the total interference penalty during the entire BH period to evaluate inter-satellite interference.

Lemma 2: The total interference penalty can be expressed as $Tr(\boldsymbol{T}^T \boldsymbol{\Omega} \boldsymbol{T})$.

Proof: Please refer to Appendix B. □

Therefore, multi-satellite interference avoidance problem can be further described as follow,

$$P_5: \underset{\boldsymbol{T}^1, \boldsymbol{T}^2, \ldots, \boldsymbol{T}^{N_s}}{\text{minimize}} \ \text{Tr}(\boldsymbol{T}^T \boldsymbol{\Omega} \boldsymbol{T})$$

$$\text{s.t. } C_1: \sum_{n=1}^{N_{slot}} t_{mn}^i = \phi_m^i, \quad m = 1, \ldots, N_v^i; i = 1, \ldots, N_s$$

$$C_2: \sum_{m=1}^{N_v^i} t_{mn}^i = 1, \quad n = 1, \ldots, N_{slot}; i = 1, \ldots, N_s$$

$$C_3: t_{mn}^i \in \{0,1\}, \quad \forall m, n, i, \quad (17)$$

where $\text{Tr}(\cdot)$ represents the trace of the matrix. C_1 constrains the number of occurrences of each valid pattern, and C_2 indicates that only one valid pattern can be executed for each satellite at each time slot.

Algorithm 3 A Heuristic Solution to P_6

Input: The number of occurrences for each valid pattern of all satellites: $[\phi_1^i, \phi_2^i, \ldots, \phi_{N_v^i}^i], i = 1, \ldots, N_s$. The number of allocated cells for all satellites: $N_i, i = 1, \ldots, N_s$;

```
Sort satellites in ascending order according to N_i: (s_1, s_2, ..., s_{N_s}) | N_{s_1} ⩽ N_{s_2} ⩽ ... ⩽ N_{s_{N_s}}.
for n = 1 to N_slot do
  Initialize the set of activated beams: B = ∅.
  for i = s_1 to s_{N_s} do
    if i = s_1 then
      Find a non-zero element φ_m^i from [φ_1^i, φ_2^i, ..., φ_{N_v^i}^i].
      Adopt the m-th valid pattern at time slot n: t_mn^i = 1.
      φ_m^i = φ_m^i − 1.
    else
      for m = 1 to N_v^i do
        if φ_m^i > 0 then
          Calculate the shortest distance η_m between the m-th valid pattern and the activated beams B according to Eq. (11).
        else
          η_m = 0.
        end if
      end for
      Find the valid pattern with the maximum distance: m* = arg max η_m.
      Adopt the m*-th valid pattern at time slot n: t_{m*n}^i = 1.
      φ_{m*}^i = φ_{m*}^i − 1.
    end if
    Update the set of activated beams B according to the adopted valid pattern.
  end for
end for
```

Output: BH time slot schedule matrix of all satellites: $\boldsymbol{T}^1, \boldsymbol{T}^2, \ldots, \boldsymbol{T}^{N_s}$.

However, $\boldsymbol{\Omega}$ may be indefinite, increasing the difficulty of solving P_5. Therefore, we modify the matrix $\boldsymbol{\Omega}$ to a positive semidefinite matrix $\hat{\boldsymbol{\Omega}}$ without changing the optimal solution of P_5. $\hat{\boldsymbol{\Omega}}$ has the form as,

$$\hat{\boldsymbol{\Omega}} = \boldsymbol{\Omega} - \lambda \mathbf{I}, \quad (18)$$

where λ is the minimum eigenvalue of $\boldsymbol{\Omega}$. As a result, the original problem P_5 can be renewed as,

$$P_6: \underset{\boldsymbol{T}^1, \boldsymbol{T}^2, \ldots, \boldsymbol{T}^{N_s}}{\text{minimize}} \ \text{Tr}(\boldsymbol{T}^T \hat{\boldsymbol{\Omega}} \boldsymbol{T})$$

$$\text{s.t. } C_1: \sum_{n=1}^{N_{slot}} t_{mn}^i = \phi_m^i, \quad m = 1, \ldots, N_v^i; i = 1, \ldots, N_s$$

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$$C_2: \sum_{m=1}^{N_v^i} t_{mn}^i = 1, \quad n = 1, \ldots, N_{slot}; i = 1, \ldots, N_s$$
$$C_3: t_{mn}^i \in \{0, 1\}, \quad \forall m, n, i. \tag{19}$$

Lemma 3: P_5 *and* P_6 *have the same optimal solution.*

Proof: Please refer to Appendix C. □

Algorithm 4 Multi-Satellite Beam Hopping Based on Load Balancing and Interference Avoidance

Input: The coordinate of each satellite $Sat(lat, lon, h)$; the coordinate of each cell covered by satellites $Cell(lat, lon)$; the traffic demand of each cell d_n; the number of satellites N_s.

Sub-problem 1: load balancing between satellites

1: $\forall i$: Calculate the set of cells covered by the i-th satellite V_i according to coordinates of satellites and cells.
2: Obtain cell-satellite allocation vector $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{N_s}$ by using Algorithm 1.

Sub-problem 2: single-satellite beam hopping

3: **for** $i = 1$ to N_s **do**
4: Calculate the number of allocated cells for the i-th satellite $N_i = sum(\boldsymbol{u}_i)$.
5: Calculate the number of activated beams for the i-th satellite M_i according to Eq. (8).
6: Obtain the number of valid patterns N_v^i and service matrix $\boldsymbol{C}_i$ according to Algorithm 2.
7: Obtain the number of times that the m-th valid BH pattern is used ϕ_m^i by solving (13).
8: **end for**

Sub-problem 3: multi-satellite interference avoidance

9: Obtain BH time slot schedule matrix of all satellites $\boldsymbol{T}^1, \boldsymbol{T}^2, \ldots, \boldsymbol{T}^{N_s}$ by Algorithm 3.

Output: The BH time slot schedule matrix $\boldsymbol{T}$.

It can be observed that P_6 corresponds to a Binary Quadratic Programming (BQP), which can be solved by the semidefinite relaxation (SDR) technique [31]. However, the complexity of directly obtaining the optimal solution by SDR is much high. As a result, this paper devises a heuristic strategy to solve P_6, as detailed in Algorithm 3. The main idea of Algorithm 3 is that at each time slot, satellites, which are sorted by number of serving cells in ascending order, select the valid pattern with maximum distance to other satellites' BH patterns. Specially, line 1 indicates sorting the satellites in ascending order according to N_i. Line 5 to line 8 represent randomly selecting one valid pattern for the first satellite, and line 10 to line 19 indicate the selection of valid pattern with maximum distance to other satellites' BH patterns, which can reduce inter-satellite interference as much as possible.

Finally, the overall multi-satellite beam hopping method based on load balancing and interference avoidance is summarized in Algorithm 4.

IV. Simulation Results and Performance Analysis

A. Simulation Parameter

In this paper, a Ka-band forward beam hopping system is simulated. Table II summarizes the simulation parameters and some parameters refer to 3GPP standard [32] and DVB-S2X standard [17]. There are 15 satellites distributed in an area with a longitude range of $[115°E, 120°E]$ and a latitude range of $[28°N, 33°N]$. The orbit height is set as 1500 km. As shown in Fig. 7, each satellite can cover 37 cells and a total of 206 cells are covered by 15 satellites. It should be noted that some cells are covered by multiple satellites since $206 < 15 * 37$. The total power of each satellite is $P_{tot} = 39$ dBW and the maximum number of beams for each satellite is equal to 4. The total bandwidth is set as 100 MHz, and the period of BHTP contains 64 time slots. Besides, the duration of time slot is set to 2 ms, which refers to DVB-S2X standard [17]. The multi-beam antenna radiation pattern refers to 3GPP TR 38.811 [32], which has the form as

$$G_t(\theta) = \begin{cases} G_m, & \theta = 0° \\ G_m \left| \dfrac{4J_1(2\pi a \sin\theta/\lambda)}{2\pi a \sin\theta/\lambda} \right|^2, & 0° < |\theta| \le 90° \end{cases} \tag{20}$$

where λ refers to the wavelength and a denotes the radius of the antenna's circular aperture. $J_1(\cdot)$ is the Bessel function of the first kind and first order. θ denotes the off-axis angle in degrees. G_m denotes the maximum gain of satellite transmit

TABLE II
Simulation Parameters

PARAMETERS	VALUES
Satellite altitude H	1500 km
Ka-band frequency f_c	20 GHz
Number of satellites N_s	15
Longitude range of satellites	$[115°E, 120°E]$
Latitude range of satellites	$[28°N, 33°N]$
Number of cells covered by each satellite N	37
Number of cells covered by all satellites N_c	206
Radius of cell R	39 km
The maximum beams for each satellite K	4
System bandwidth B	100 MHz
Total satellite power P_{tot}	39 dBW
The radius of the antennas aperture a	0.15 m
3 dB beamwidth of satellite beams	3°
Maximum transmit antenna gain G_m	35.9 dBi
Terminal receiving antenna gain G_r	0 dBi
Noise temperature T_{rx}	300 K
Time slot duration T_{slot}	2 ms (Refer to DVB-S2X [17])
The period of BHTP T_H	$64 \times T_{slot}$

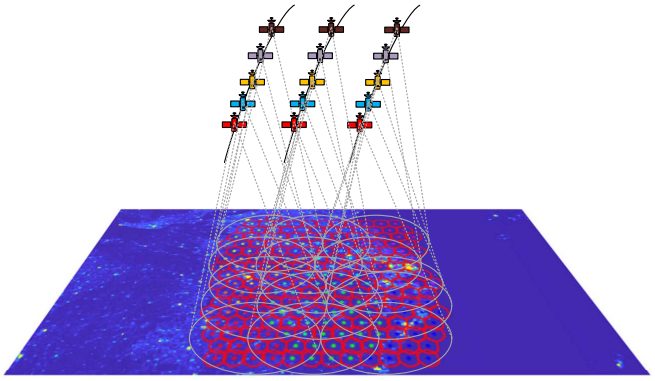


Fig. 7. Illustration of the coverage for 15 NGSO satellites.

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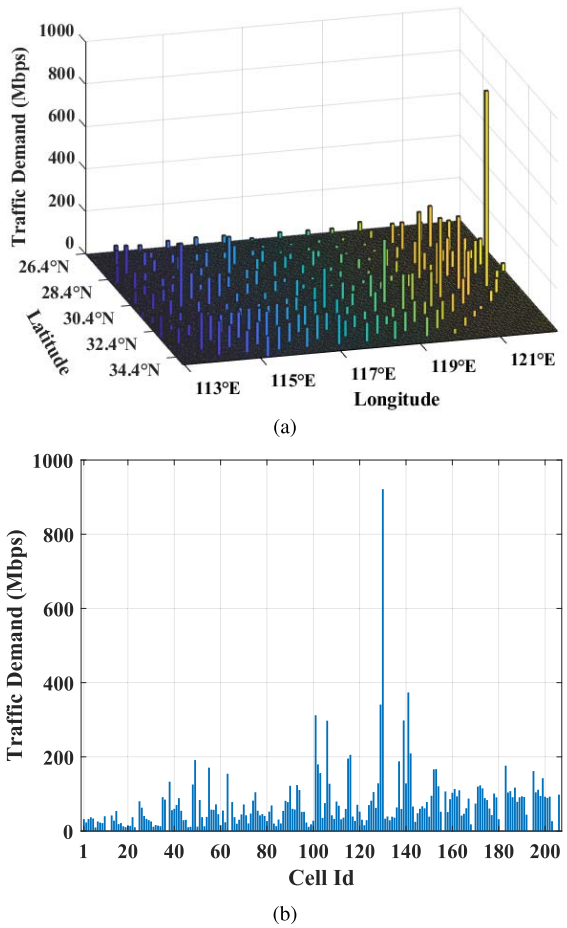


Fig. 8. Illustration of the traffic of each cell.

antenna, which has a form as $G_m = 10 * \log_{10}[(\frac{2\pi a}{\lambda})^2]$. Specially, the user receiving antenna defaults to an omnidirectional antenna, thus the receiving gain is set to 0 dBi.

The aggregate traffic demand of each cell is mainly obtained according to the world population distribution [33]. Fig. 8 (a) illustrates the traffic demand of 206 cells, where the WGS-84 geodetic coordinates are transformed into geostationary rectangular coordinates. It can be seen from Fig. 8 (b) that the traffic demand shows serious non-uniformity. Especially, the 130-th cell has huge demand due to the ultra-dense population distribution.

In this paper, we compare the proposed method with seven different BH schemes as follows,

1) The Random Beam Hopping (R-BH): The R-BH method means that each satellite randomly selects 4 cells from 37 cells for service at each time slot.

2) The Periodic Beam Hopping (P-BH): The P-BH method takes a periodic beam hopping, which means each satellite will serve the covered cells periodically. In this simulation, each satellite covers 37 cells and generates 4 beams. Thus, the duration of each period is about $37/4 \approx 9$ time slots.

3) The Greedy-Based Beam Hopping (G-BH): In G-BH algorithm [34], each satellite obtains the traffic demand of 37 cells that it covers and greedily severs the top 4 cells with the most traffic demand at each time slot.

4) Load Balancing and Greedy-Based Beam Hopping (LB-G-BH): In LB-G-BH algorithm [34], load balancing is implemented by solving P_2. Then, each satellite obtains the traffic demand of allocated cells, and greedily severs the top 4 cells with the most traffic demand at each time slot.

5) Single-Satellite Independent Beam Hopping (SA-BH): In SA-BH algorithm [23], each satellite obtains the traffic demand of 37 cells that it covers, and solves BH pattern design problem P_4 independently to get its own BHTP. In other words, SA-BH does not consider load balancing and interference avoidance between satellites.

6) Load Balancing and Single-Satellite Independent Beam Hopping (LB-SA-BH):

LB-SA-BH algorithm solves the load balancing problem P_2 and the BH pattern design problem P_4 [23], but dose not consider the interference avoidance problem P_6.

7) Load Balancing and Genetic Algorithm Beam Hopping (LB-GA-BH):

In LB-GA-BH algorithm, load balancing is firstly implemented by solving P_2. Then, the genetic algorithm [35] is adopted in each satellite to get its own BHTP where the population size is set as 2000.

B. Performance Analysis

1) Performance of Load Balancing:

In order to balance the non-uniform traffic demand to each satellite, load balancing problem P_2 is solved. Fig. 9 (a) shows the satellite-cell assignment results after load balancing. We can observe that some satellites will serve lots of low-demand cells (e.g. satellite 6 and 11), while some satellites (e.g. satellite 13 and 14) only need to serve a few hotspot cells. As a result, the traffic load gap between different satellites can be effectively reduced, which can greatly improve the resource utilization of NGSO satellite constellation. In order to show the relative load relationship between satellites, we define the normalized load, which has the form as

$$\hat{S}_i = \frac{S_i}{\sum_{i=1}^{15} S_i}, \tag{21}$$

where $\hat{S}_i$ denotes the normalized traffic demand that should be served by satellite i, and S_i is the traffic load of satellite i, which has been defined in (6).

Fig. 9 (b) illustrates the normalized traffic demand of each satellite before and after load balancing. The blue dotted line represents the optimal load balancing strategy by using MOSEK to solve P_2, while the yellow line indicates a suboptimal solution provided by Algorithm 1. Assume that cells are allocated to satellites according to the maximum elevation angle before load balancing. It can be observed that due to uneven traffic distribution, the load gap between satellites has reached 0.091 before load balancing. Particularly, satellite 13 and 14 have much traffic because of closest to traffic-intensive area. However, the load gap has dropped to 0.025 with the help of Algorithm 1, indicating that the load variance is reduced by about 72.5%. Specially, the load on satellite 13 and 14 has been effectively lightened by about

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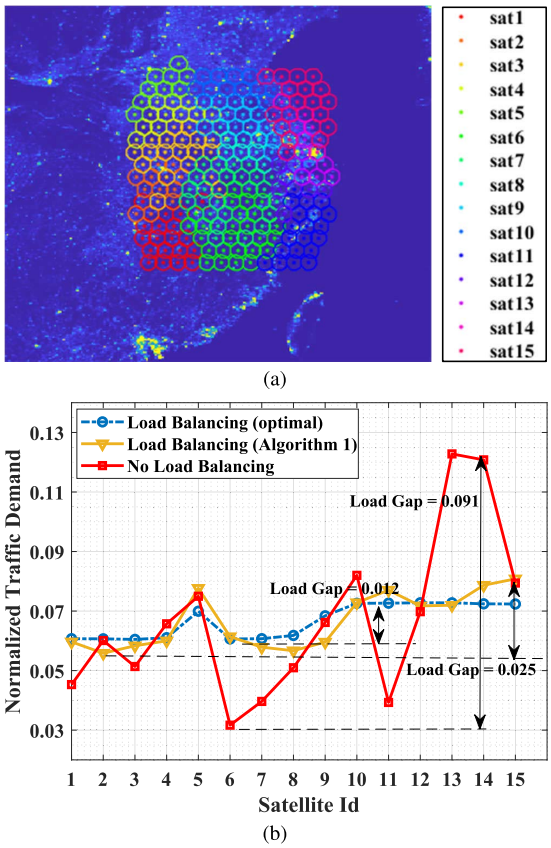


Fig. 9. Results of load balancing, (a) the satellite-cell assignment results after load balancing and (b) the normalized traffic demand of 15 satellites before and after load balancing.

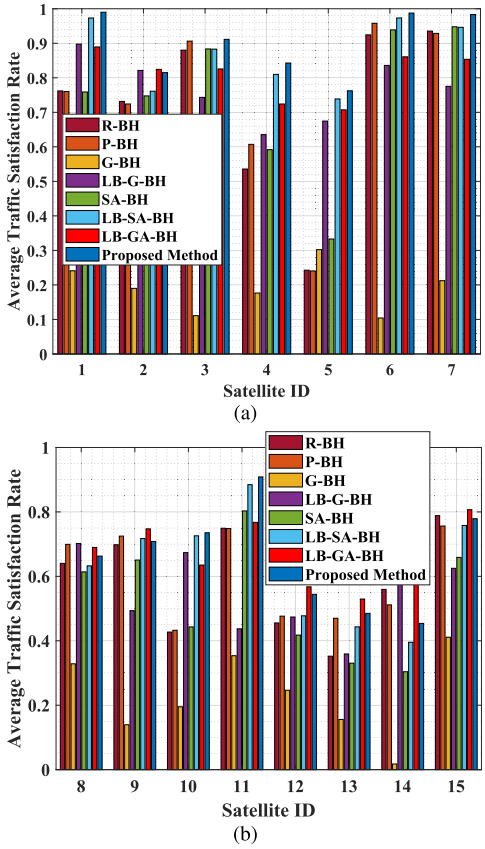


Fig. 10. Average traffic satisfaction rate of 15 satellites among different beam hopping algorithms.

40% since fewer cells are assigned to them. Compared with the optimal load balancing scheme, Algorithm 1 slightly losses around 10% performance but the computational complexity is greatly reduced.

2) Performance of Traffic Satisfaction Rate:

In this paper, the traffic satisfaction rate is defined to evaluate the performance of different beam hopping algorithms, which has the form as

$$\beta_n = \frac{\bar{R}_n}{D_n}, \tag{22}$$

where β_n represents the ratio of the offered traffic to the demand traffic in term of cell n; $\bar{R}_n = \min\{R_n, D_n\}$ and R_n, D_n are defined in (5). Obviously, $\beta_n \leqslant 1$ and the larger the value is, the more satisfied the cell is. For visualization of the results, we further define the average traffic satisfaction rate of each satellite as $\hat{\beta}_i = \frac{\sum_{n=1}^{N_i} \beta_n}{N_i}$, where N_i denotes the number of cells that the i-th satellite actually serves.

The average traffic satisfaction rate of 15 satellites are shown in Fig. 10. It can be seen that our proposed method outperforms other algorithms for most satellites, since the proposed method not only considers on-demand transmission, but also effectively avoids intra-satellite interference and inter-satellite interference. Besides, the traffic satisfaction rate of some satellites are close to 1, which means that the demands of these cells are almost completely satisfied in our method. Since P-BH and R-BH do not focus on on-demand transmission, many cells cannot be effectively satisfied, resulting in the satisfaction rate far less than 1. Although G-BH preferentially selects cells with high demand for transmission, it fails to serve cells with low traffic, resulting in 0 satisfaction rate in those cells. Furthermore, SA-BH and LB-GA-BH only optimize each satellite independently, ignoring the interference avoidance between satellites; as a result, they perform worse than the proposed method. In addition, it is noted that whether G-BH or SA-BH, the performance has been improved after load balancing. There are two main reasons for this phenomenon. On the one hand, the corresponding demand of each satellite is balanced, and the case of resource waste caused by light load of satellites is avoided as much as possible. On the other hand, the load balancing algorithm ensures that the cells corresponding to each satellite will not overlap, so there will be no situation that a certain cell is served by several satellites at the same time, which avoids inter-satellite interference to a certain extent. Furthermore, it can be observed that all algorithms perform poorly on satellite 13 because cell 130 belonging to satellite 13 has huge demand and the limited satellite resources are far from meeting the needs of this cell.

3) Performance of Unmet and Unused Capacity:

In order to better illustrate the effectiveness of resource allocation, different algorithms are evaluated in terms of unmet capacity and unused capacity for each satellite. The unmet

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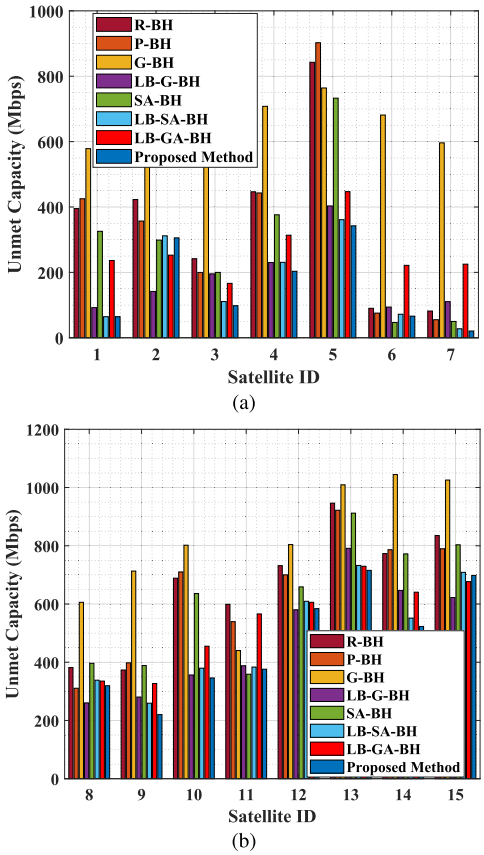


Fig. 11. Unmet capacity of 15 satellites among different beam hopping algorithms.

capacity corresponds to the amount of capacity which do not meet the actual demand, and is defined as $C^i_{unmet} = \sum_{n=1}^{N_i} (D_n - R_n)^+ / T_H$, where $(x)^+ = \max(0, x)$. Besides, the unused capacity represents the offered capacity which exceeds the demand, and has the form as $C^i_{unused} = \sum_{n=1}^{N_i} (R_n - D_n)^+ / T_H$.

Fig. 11 shows the unmet capacity of different algorithms. We can observe that the proposed algorithm has the smallest unmet capacity, which indicates that our method outperforms others in terms of on-demand services. Specially, the unmet capacity is reduced by 37.8%, 35.9%, 56.1%, 6.0%, 29.8%, 5.0% and 9.2% compared with R-BH, P-BH, G-BH, LB-G-BH, SA-BH, LB-SA-BH and LB-GA-BH. Therefore, our method can more effectively match limited satellite resources to non-uniform traffic demand. It should be noted that satellite 13 and 14 have much unmet capacity under all algorithms since their huge demands far exceeds the capabilities of the satellites. Nonetheless, our algorithm serves these high-demand cells as much as possible compared to other algorithms due to the smaller unmet capacity. R-BH and P-BH can not well meet the traffic of each cell since they don't focus on on-demand services. In addition, G-BH exhibits poor performance since it struggles in meeting the demand of low-demand cells because of the priority to serve high-demand cells. Although LB-SA-BH and LB-GA-BH can meet low-demand cells, the inter-satellite interference has not been effectively suppressed; as a result, high-demand cells cannot be well satisfied due to some inter-satellite interference.

TABLE III
UNMET CAPACITY, UNUSED CAPACITY AND AVERAGE TRAFFIC SATISFACTION RATE RESULTS OF EIGHT ALGORITHMS

Algorithm	Total Unmet Capacity (Mbps)	Total Unused Capacity (Mbps)	Average Satisfaction Rate (%)
R-BH	7850	1908	71.7
P-BH	7615	2018	72.4
G-BH	11124	0	21.9
LB-G-BH	5192	337	68.9
SA-BH	6956	657	71.5
LB-SA-BH	5141	1220	79.8
LB-GA-BH	6199	1780	76.9
Proposed Method	**4883**	**1485**	**81.4**

For convenience, Table III summarizes the total unmet capacity, the total unused capacity and the average traffic satisfaction rate of all cells under different methods. Particularly, compared with R-BH, P-BH, G-BH, LB-G-BH, SA-BH, LB-SA-BH and LB-GA-BH, the average traffic satisfaction rate of the proposed method can be improved by 13.5%, 12.4%, 271.7%, 18.1%, 13.8%, 2.0% and 5.9% respectively. The total unused capacity represents the amount of wasted capacity for the whole system. Although the proposed method has a little wasted capacity on some satellites, it can not directly account for inefficiency. For instance, G-BH has 0 unused capacity while it shows the smallest average satisfaction rate. Actually, in order to serve the low-demand cells for fairness, it will inevitably lead to a certain waste of resources. Therefore, the service efficiency of the whole system is mainly reflected in the unmet capacity.

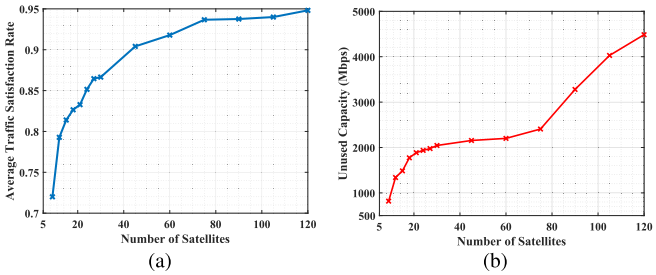
C. Computational Complexity and Scalability Analysis

Firstly, we analyze the complexity of Algorithm 1 for solving sub-problem 1 (i.e., load balancing problem). Line 1 for sorting cells in descending order according to d_n has the complexity of $O(N_c \log(N_c))$. Besides, line 2 to line 7 for assigning each cell to the visible satellite with minimum traffic load has the complexity of $O(N_c N_s)$. Therefore, the overall complexity of Algorithm 1 is $O(N_c(N_s + \log(N_c))$.

Secondly, we analyze the complexity of the solution to sub-problem 1 (i.e., single-satellite beam hopping pattern design problem). As for satellite i, selecting spatial isolation patterns as shown in Algorithm 2 has the complexity of $O(C^{M_i}_{N_i})$. Then, P_4 is solved using branch and bound method by MOSEK solver, which has the the complexity of $O(2^{N^i_v})$ in worst-case. Thus, the overall complexity of sub-problem 2 is $O(\sum_{i=1}^{N_s} (C^{M_i}_{N_i} + 2^{N^i_v}))$.

Thirdly, we analyze the complexity of Algorithm 3 for solving sub-problem 3 (i.e., interference avoidance problem). Line 1 for sorting satellites in ascending order according to N_i has the complexity of $O(N_s \log(N_s))$. Line 2 to line 23 for selecting the valid pattern with maximum distance has the

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(a) (b)

Fig. 12. Performance of the proposed algorithm for different number of satellites in terms of (a) average traffic satisfaction rate and (b) unused capacity.

complexity of $O(N_{slot}\sum_{i=1}^{N_s} N_v^i)$. As a result, the overall complexity of sub-problem 3 is $O(N_s\log(N_s)) + N_{slot}\sum_{i=1}^{N_s} N_v^i)$.

In summary, the computational complexity of whole algorithm can be expressed as $O(N_c(N_s + \log(N_c) + \sum_{i=1}^{N_s}(C_{N_i}^{M_i} + 2^{N_v^i}) + N_s\log(N_s) + N_{slot}\sum_{i=1}^{N_s} N_v^i) \approx O(N_s \max_{i=1,2,\ldots N_s}\{C_{N_i}^{M_i} + 2^{N_v^i} + N_{slot}N_v^i + N_c\})$. In a dense network, each satellite only covers about dozens of cells, and the number of activated beams is much small due to power limitations. Thus, $C_{N_i}^{M_i}$ and N_v^i will not be so large and the complexity can be acceptable. In addition, it can be observed that the overall computational complexity is linear with the number of satellites N_s, which means that the proposed method has good scalability when the number of satellites is increased.

Fig. 12 shows the performance of the proposed algorithm for different number of satellites in terms of average traffic satisfaction rate and unused capacity. The simulation settings are the same as Table II and the number of satellite is set as 9 to 120. It can be noted that as the number of satellites increases, the average satisfaction rate tends to 1 due to the excessive satellite resources. However, superfluous satellites will cause serious waste of resources, which is reflected in unused capacity. As shown in Fig. 12 (b), the unused capacity rises sharply when the number of satellites exceeds 80. Therefore, the scale of satellite should not be too large, in order to make a compromise on service performance and resource waste.

V. Conclusion

This paper studied the problem of NGSO multi-satellite beam hopping to match the limited satellite resources to the non-uniform traffic demand. At first, the optimization problem for the multi-satellite beam hopping based on full frequency reuse is formulated. Being aware of the NP-hardness of the optimization problem, this paper decomposed the original problem into three sub-problems. On the one hand, the multiple coverage characteristics of the NGSO constellation is used to balance the demand between satellites and ensure the maximum utilization of resources; on the other hand, by selecting spatially isolated beam hopping patterns and making BH pattern scheduling, we can achieve on-demand service and interference avoidance. Finally, simulations have proven that the proposed method has better performance in terms of traffic satisfaction rate and unmet capacity than existing beam hopping approaches.

Appendix A
Proof of Lemma 1

Proof: We construct a reduction from mixed integer linear programming (MILP), a well-known NP-hard problem, to a special instance of our optimization problem P_0. MILP can be formulated as

$$\max_{x\in\mathcal{S}} d^T x, \tag{23}$$

where $d \in \mathbb{Q}^n$ and $\mathcal{S} = \{x \in \mathbb{R}^n \mid Ax \le b\} \cap (\mathbb{Z}^r \times \mathbb{R}^{n-r})$ for some nonnegative integer r.

We consider a special instance of P_0, where $P_k = P_{sat}/K$ and the co-frequency interference is ignored. As a result, C_2 in P_0 can be expressed as

$$\begin{aligned} R_n &= \sum_{t=1}^{N_{slot}} C_n^t \\ &= \sum_{t=1}^{N_{slot}} \sum_{i,n\in V_i} x_{i,m_i}^t BT_{slot}\log_2(1+\frac{P_{sat}}{k_B T_{rx} BK}) \\ &= \sum_{t=1}^{N_{slot}} \sum_{i,n\in V_i} c x_{i,m_i}^t, \end{aligned} \tag{24}$$

where $c = BT_{slot}\log_2(1 + P_{sat}/k_B T_{rx} BK)$ is a constant and V_i denotes the set of cells covered by the i-th satellite. m_i indicates that the m_i-th cell covered by the i-th satellite corresponds to cell n. We define an auxiliary variable $\xi \triangleq \min_n \frac{R_n}{D_n}$ and P_0 can be renewed as follow,

$$\begin{aligned} P_0': \underset{\boldsymbol{X}_1,\boldsymbol{X}_2,\ldots,\boldsymbol{X}_{N_s},\xi}{\text{maxmize}}\ & \xi \\ \text{s.t. } C_1: & \sum_{n=1}^{N} x_{i,n}^t \leqslant K, \quad t=1,\ldots,N_{slot}; i=1\ldots,N_s, \\ C_2: & D_n\xi - \sum_{t=1}^{N_{slot}} \sum_{i,n\in V_i} c x_{i,m_i}^t \leqslant 0, \quad n=1,\ldots,N_c, \\ C_3: & x_{i,n}^t \in \{0,1\}, \quad i=1\ldots,N_s; n=1,\ldots,N. \end{aligned} \tag{25}$$

It can be seen that P_0' is a MILP due to linear constraints and linear objective. Since P_0' is a special and easy instance of P_0 and P_0' is NP-hard, P_0 is NP-hard. This completes the proof. □

Appendix B
Proof of Lemma 2

Proof: We let $\boldsymbol{J} = \boldsymbol{T}^T\boldsymbol{\Omega}\boldsymbol{T}$. According to $\boldsymbol{T} = [\boldsymbol{T}^1;\boldsymbol{T}^2;\ldots;\boldsymbol{T}^{N_s}]$, the k-th diagonal element of $\boldsymbol{J}$ can be written as

$$\begin{aligned} \boldsymbol{J}_{k,k} &= [\boldsymbol{T}^1(:,k);\ldots;\boldsymbol{T}^{N_s}(:,k)]^T\boldsymbol{\Omega}[\boldsymbol{T}^1(:,k);\ldots;\boldsymbol{T}^{N_s}(:,k)] \\ &= \sum_{m=1,m\neq n}^{N_s}\sum_{n=1}^{N_s} (\boldsymbol{T}^m(:,k))^T\Omega_{m,n}\boldsymbol{T}^n(:,k). \end{aligned} \tag{26}$$

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It can be observed that $(\boldsymbol{T}^m(:,k))^T\Omega_{m,n}\boldsymbol{T}^n(:,k)$ represents the interference penalty between satellite m and satellite n at time slot k. Therefore, $\boldsymbol{J}_{k,k}$ indicates the total interference penalty among all satellites at k-th time slot. As a result, $\mathrm{Tr}(\boldsymbol{T}^T\boldsymbol{\Omega}\boldsymbol{T}) = \mathrm{Tr}(\boldsymbol{J}) = \sum_{k=1}^{N_{slot}} \boldsymbol{J}_{k,k}$ can express the total interference penalty during the entire BH period. This completes the proof. □

APPENDIX C
PROOF OF LEMMA 3

Proof: Assuming that $\boldsymbol{T}^*$ is the optimal solution of P_6, we have the form as

$$\mathrm{Tr}(\boldsymbol{T}^{*T}\hat{\boldsymbol{\Omega}}\boldsymbol{T}^*) \geqslant \mathrm{Tr}(\boldsymbol{T}^T\hat{\boldsymbol{\Omega}}\boldsymbol{T}), \forall \boldsymbol{T}. \tag{27}$$

According to $\hat{\boldsymbol{\Omega}} = \boldsymbol{\Omega} - \lambda\mathbf{I}$, we have the following form as,

$$\mathrm{Tr}(\boldsymbol{T}^T\boldsymbol{\Omega}\boldsymbol{T}) = \mathrm{Tr}(\boldsymbol{T}^T\hat{\boldsymbol{\Omega}}\boldsymbol{T}) + \lambda\mathrm{Tr}(\boldsymbol{T}^T\boldsymbol{T}). \tag{28}$$

Considering $\boldsymbol{T} = [\boldsymbol{T}^1; \boldsymbol{T}^2; \ldots; \boldsymbol{T}^{N_s}]$, we can obtain,

$$\begin{aligned}\mathrm{Tr}(\boldsymbol{T}^T\boldsymbol{T}) &= \mathrm{Tr}([\boldsymbol{T}^1; \boldsymbol{T}^2; \ldots; \boldsymbol{T}^{N_s}]^T[\boldsymbol{T}^1; \boldsymbol{T}^2; \ldots; \boldsymbol{T}^{N_s}]) \\ &= \mathrm{Tr}({\boldsymbol{T}^1}^T\boldsymbol{T}^1) + \ldots + \mathrm{Tr}({\boldsymbol{T}^{N_s}}^T\boldsymbol{T}^{N_s}) \\ &= \sum_{n=1}^{N_{slot}}\sum_{m=1}^{N_v^1}(t_{mn}^1)^2 + \ldots + \sum_{n=1}^{N_{slot}}\sum_{m=1}^{N_v^{N_s}}(t_{mn}^{N_s})^2. \end{aligned} \tag{29}$$

According to the fact $(t_{mn}^i)^2 = t_{mn}^i$ and C_2 in P_6, Eq. (29) can be further expressed as follow,

$$\begin{aligned}\mathrm{Tr}(\boldsymbol{T}^T\boldsymbol{T}) &= \sum_{n=1}^{N_{slot}}\sum_{m=1}^{N_v^1}(t_{mn}^1)^2 + \ldots + \sum_{n=1}^{N_{slot}}\sum_{m=1}^{N_v^{N_s}}(t_{mn}^{N_s})^2 \\ &= \sum_{n=1}^{N_{slot}}\sum_{m=1}^{N_v^1}(t_{mn}^1) + \ldots + \sum_{n=1}^{N_{slot}}\sum_{m=1}^{N_v^{N_s}}(t_{mn}^{N_s}) \\ &= \sum_{n=1}^{N_{slot}} 1 + \ldots + \sum_{n=1}^{N_{slot}} 1 \\ &= N_{slot}N_s. \end{aligned} \tag{30}$$

Thus, we can obtain $\mathrm{Tr}(\boldsymbol{T}^{*T}\boldsymbol{\Omega}\boldsymbol{T}^*) = \mathrm{Tr}(\boldsymbol{T}^{*T}\hat{\boldsymbol{\Omega}}\boldsymbol{T}^*) + \lambda N_{slot}N_s \geqslant \mathrm{Tr}(\boldsymbol{T}^T\hat{\boldsymbol{\Omega}}\boldsymbol{T}) + \lambda N_{slot}N_s = \mathrm{Tr}(\boldsymbol{T}^T\boldsymbol{\Omega}\boldsymbol{T})$, which shows that $\boldsymbol{T}^*$ is the optimal solution of P_5. This completes the proof. □

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